2.5.3.2 How to read the hazard maps – what the hazard scenario means

When looking at the hazard map, the rate class (365 m² year-1 km-2) means that for an area of 1 km² (i.e. 106 m2) that belongs to the top 10% of the landscape that is the most susceptible to shallow landslide initiations in western Rwanda, 365 m² will be affected by the initiation of a slope failure every year. One can say that in ~2700 years; each pixel of this hazard area has the probability of being affected by the initiation of landslides. Note that this scenario is only here for the initiation of the landsides (i.e., the source area of the landslide) and does not consider the total area of the landslides (i.e. the source area + the impact zones – Figure 3 b). For every m² of new landslide initiation, 5 m² of landslide impacts are to be assumed (see description of inventory in Step 2).

We can therefore say that every year, for an area of 1 km2, 6*365 m² will be impacted by landslides. In other words, this represents 0.2 % of the area that are impacted every year. Knowing that it is based on an inventory from satellite images from which the identification of the landslides is always an underestimation (see earlier sections), we can fairly say that, overall, in the highest hazard class, every year about 1% of the land is impacted by the occurrence of new landslides.

Table 7: **Areas per landslide hazard rate class per site**

Hazard rate class (m^2/year/km^2	Bishenyi		Rwabayanga		Rusizi		Rwandex-Magerwa	
	Area (km^2)	Percentage of total area	Area (km^2)	Percentage of total area	Area (km^2)	Percentage of total area	Area (km^2)	Percentage of total area
1	21.06	44.6%	5.34	68.3%	15.98	75.9%	6.77	69.7%
18	19.54	41.3%	1.84	23.5%	4.40	20.9%	2.19	22.5%
90	5.96	12.6%	0.59	7.5%	0.66	3.1%	0.68	7.0%
365	0.71	1.5%	0.05	0.6%	0.02	0.1%	0.08	0.8%

The hazard class of 365 m²/year/km2 that corresponds to the highest value in western Rwanda, corresponds to a very high hazard for the Lake Kivu region in general (Depicker et al., 2020a). When looking at the global scale, the highest hazard that we find in this study is relatively moderate (Larsen and Montgomery, 2010; Broeckx et al., 2020). However, such an assessment must be considered with care since environmental conditions of other regions can be highly different.

In Step 1, we explained that, in the Lake Kivu region, clusters of landslides (sometimes with thousands of slope failures) can occur and that their spatio-temporal distribution is random on a few years basis since they are associated with extreme convective rainfall events. This randomness must be kept in mind and an area that is not highlighted as hazard-prone can still be impacted by dramatic landslide events.